Sunk ROBOTICS

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Sunk Robotics JENA Float

Movement:

The JENA Float utilizes a buoyancy engine to control its vertical position within the water column. Utilizing a piston powered by a motor equipped with a precise rotary encoder, the JENA Float is able to change its density to rise or sink, at varying speeds. A PID controller, used to determine the ideal piston location, and a proportional controller, in combination with the piston head's high friction with the shaft, and a natural dead band below 30% speed, allows for accurate and efficient confluence with the desired vertical position.

Safety:

At Sunk Robotics safety is our #1 priority. In order to accommodate this, JENA operates by pulling a vacuum rather than pressurizing the cylinder, ensuring the pressure within the JENA Float never rises above that of the surrounding water. In the rare event of improper depressurization, the JENA Float is equipped with a 2.5cm rubber stopper designed to alleviate abnormal levels of internal pressure.

Communication:

The JENA Float is equipped with an external antenna that protrudes above the water's surface, allowing the JENA Float to communicate via WiFi and Websockets to an access point created on the client's laptop on shore.

Over this connection, the JENA Float can receive and interpret the following commands:

- **profile** - causes the float to complete a full vertical profile
- **get_data** - returns collected depth data
- **get_pressure** - returns collected pressure data
- **break** - cancels communication with the float and causes the float to return to it's awaiting communication mode
- **set_target_depth -** sets the target depth of the JENA Float, to be reached and held upon the running of the profile command.
- **set_hold_time** - specifies the time that the JENA Float should spend collecting depth and pressure data at the desired depth.

The depth or pressure data is returned in an array where each number represents 5 seconds elapsing in float time and each value is the depth or pressure at that point. Due it its convenient structure, the data is easily available for clients to use for their desired purposes.

Sunk Robotics' JENA